

LAWRENCE, MA, USA

WMO: 744904

Lat: 42.717N

Lon: 71.124W

Elev: 149

StdP: 14.62

Time Zone: -5.00 (NAE)

Period: 97-19

WBAN: 94723

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.6	9.6	-11.8	2.9	8.8	-7.6	3.7	12.9	24.4	26.4	21.6	26.8	8.7	300	0.503

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.5	90.8	73.1	88.1	71.8	84.9	70.2	75.9	86.1	74.4	83.6	72.9	81.3	10.0	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.7	121.9	80.5	71.8	118.1	79.7	70.2	111.6	78.1	39.3	86.0	37.9	83.5	36.5	81.4	81.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.5	18.3	16.3	DB	-1.0	95.5	5.3	3.0	-4.8	97.7	-7.9	99.4	-10.8	101.1	-14.7	103.4	(4)
(5)			WB	-2.4	78.8	4.9	1.2	-6.0	79.7	-8.9	80.4	-11.7	81.1	-15.2	82.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.6	27.7	29.7	36.9	48.3	58.4	67.1	73.2	71.9	64.6	52.8	42.6	33.1	(6)	
	DBStd	17.83	10.55	9.33	9.46	8.34	7.78	7.25	5.74	5.62	7.17	7.79	8.44	8.98	(7)	
	HDD50	2660	693	569	423	127	11	0	0	0	1	58	250	528	(8)	
	HDD65	5975	1157	987	874	505	238	60	6	9	93	387	671	988	(9)	
	CDD50	2893	2	2	15	77	271	512	720	677	439	144	30	4	(10)	
	CDD65	732	0	0	1	5	32	122	261	221	82	8	0	0	(11)	
	CDH74	6340	0	1	16	85	389	1101	2313	1780	600	55	0	0	(12)	
	CDH80	2116	0	0	4	26	122	370	841	576	173	4	0	0	(13)	
	Wind	WSAvg	7.0	7.9	8.0	8.5	8.0	7.0	6.5	6.0	5.6	5.8	6.8	7.1	7.4	(14)
	Precipitation	PrecAvg	42.6	3.6	3.0	3.7	3.8	3.6	3.5	3.2	3.1	3.7	3.8	4.0	3.7	(15)
(16) Precipitation	PrecMax	60.6	12.9	8.6	11.3	12.1	7.9	12.1	7.1	9.1	9.2	14.6	9.7	8.6	(16)	
	PrecMin	28.2	0.6	0.5	0.5	0.7	0.9	0.5	0.7	0.7	0.7	0.4	0.8	0.0	(17)	
	PrecStd	6.9	2.8	1.6	2.0	2.0	1.7	2.6	1.7	1.8	2.2	2.7	2.0	2.0	(18)	
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.3	61.1	74.7	83.7	88.7	92.4	94.7	93.5	90.9	79.4	69.8	63.7	(19)	
		MCWB	56.0	52.5	59.5	62.7	67.7	74.0	75.0	74.7	73.0	65.5	60.0	57.7	(20)	
	2%	DB	53.0	53.1	63.5	74.1	83.4	88.4	90.9	89.7	84.6	74.6	64.3	55.3	(21)	
		MCWB	47.0	46.1	52.4	57.8	65.9	72.5	73.6	72.8	70.2	64.6	57.3	50.3	(22)	
	5%	DB	46.9	47.5	56.0	68.3	78.2	84.1	88.0	86.0	80.7	70.2	60.8	50.9	(23)	
		MCWB	41.8	40.9	46.5	54.7	62.7	69.3	72.3	71.1	68.4	62.3	53.6	46.3	(24)	
	10%	DB	41.8	42.8	50.3	63.1	73.0	80.8	84.5	82.7	76.7	65.6	55.9	46.2	(25)	
		MCWB	36.7	36.9	43.1	51.7	60.7	67.6	70.9	69.8	66.6	58.5	50.5	41.8	(26)	
	0.4%	WB	57.1	56.4	61.2	65.3	71.1	76.8	78.3	77.4	75.3	70.3	63.4	58.3	(27)	
		MCDB	60.0	60.1	71.9	76.3	85.0	88.2	90.4	89.0	84.9	75.5	67.8	62.1	(28)	
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	47.8	47.7	54.7	61.0	68.3	74.6	76.0	75.4	72.8	66.5	59.4	52.0	(29)	
		MCDB	51.4	52.2	61.9	72.3	79.4	84.4	86.6	84.5	80.7	72.2	63.7	55.0	(30)	
	5%	WB	42.2	41.3	48.2	57.3	65.4	72.3	74.4	74.0	71.0	63.0	55.0	47.0	(31)	
		MCDB	47.1	46.4	53.7	65.9	74.6	81.0	84.1	81.8	77.6	69.0	58.9	49.8	(32)	
	10%	WB	37.2	37.5	43.6	52.6	62.4	69.9	72.8	72.4	69.0	59.6	51.1	42.4	(33)	
		MCDB	41.0	41.2	49.6	61.0	70.6	77.5	81.7	79.8	75.0	64.4	55.5	46.1	(34)	
(36) Mean Daily Temperature Range	5% DB	MDBR	14.5	16.0	16.4	18.9	19.3	18.5	18.5	18.4	18.6	17.4	15.7	14.0	(35)	
		MCDBR	20.2	21.5	24.3	27.9	27.6	23.5	22.0	21.6	22.0	22.2	21.0	19.5	(36)	
		MCWBR	17.0	16.3	15.8	15.3	13.6	11.1	9.0	8.7	10.7	13.5	15.6	16.9	(37)	
	5% WB	MCDBR	19.9	19.8	22.5	25.0	24.2	20.8	19.1	18.1	17.7	19.9	17.7	19.1	(38)	
		MCWBR	17.8	16.6	17.1	16.5	13.5	11.2	9.3	8.9	10.6	14.3	15.5	18.2	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.285	0.295	0.320	0.363	0.399	0.441	0.461	0.428	0.371	0.331	0.307	0.289	(40)	
	taud		2.489	2.479	2.484	2.400	2.338	2.249	2.203	2.277	2.421	2.542	2.568	2.561	(41)	
	Ebn at Noon		275	290	294	287	278	265	258	263	272	273	265	264	(42)	
	Edh at Noon		23	27	30	36	39	43	45	40	32	25	21	20	(43)	
(44) All-Sky Solar Radiation	RadAvg		554	829	1152	1469	1630	1760	1859	1676	1348	882	599	452	(44)	
	RadStd		35	66	77	158	188	165	117	78	89	78	50	49	(45)	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+247	+153
(47) Regional (75 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+164	+68

Nomenclature: See separate page